

LSC Canonical Freeze Core

LSC Canonical Core

Status: Step 1 canonicalization for the Pre-BEST-2 freeze program
Scope: LSC / SC neutrino phenomenology only
Current active reference state: LSC 6.3.0 Unified BEST-2 Continuation
Primary identity: Zenodo concept DOI [10.5281/zenodo.19780615](https://doi.org/10.5281/zenodo.19780615)

1. Canonical Status

LSC is frozen as a constrained phenomenological validation framework.

It is not frozen as:

- confirmed physics,
- a discovery claim,
- a sterile-neutrino exclusion,
- a PBH/curvature theory,
- a new particle model,
- an AI/meta-analysis framework.

The active scientific question is:

Can a pre-registered detector-response model with scalar trace plus traceless anisotropic components structure gallium-source residuals better than null, normalization-only, cross-section/systematics, sterile-neutrino, and detector-systematics alternatives, while surviving external vetoes?

2. Active Canonical Concepts

2.1 Gallium-source residual analysis

The active domain is BEST/Gallium source-experiment residual analysis, including BEST, GALLEX and SAGE where data are available.

Canonical role:

- define rate-ratio residuals;
- compare against required baselines;
- avoid claiming new physics before validation;
- preserve missing-data limits.

2.2 Detector-response framing

The active LSC object is a detector-response template, not a fundamental field.

Canonical role:

- separate detector response from propagation speculation;

- express residual templates in terms of analysis bins or detector zones;
- treat detector systematics as an active competing explanation.

2.3 Trace / traceless split

The scalar trace response and traceless directional anisotropy split is active and canonical because it reduces ambiguity between:

- global normalization shifts;
- cross-section/source-strength shifts;
- detector-zone effects;
- directional or orientation-dependent structure.

Audit consolidation:

- Kimi: freeze trace/traceless formalism as the cleanest structural element.
- Gemini: freeze with downgraded claims because directional tensor components are currently underconstrained.
- DeepSeek: detector/systematics can mimic trace and traceless components unless explicitly broken by metadata and calibration.
- Claude/physics audit: trace/traceless separation is salvageable only if anisotropy is pre-defined rather than freely fitted.

2.4 Frozen prediction protocol

Before BEST-2-style fitting, the following must be frozen:

- equations used;
- parameter list;
- priors or bounds;
- datasets;
- baseline models;
- success criteria;
- failure criteria;
- treatment of missing covariance;
- external veto rules.

Any change after seeing fit output requires a new version and cannot be treated as the same freeze.

2.5 Required baseline hierarchy

The active canonical baseline hierarchy is:

- M0: null / published baseline;
- M1: scalar normalization only;
- M2: cross-section/systematics only;
- M3: sterile-neutrino benchmark;
- M4: detector-systematics benchmark;

- M5: LSC trace-only;
- M6: LSC trace plus traceless anisotropy.

LSC can only claim relative support if M5/M6 improve predictive performance after model-complexity penalties and after comparison to M0-M4.

2.6 Leave-one-out and cross-experiment validation

Anchor fits are not validation.

Active validation requires:

- leave-one-out stability;
- BEST inner-to-outer and outer-to-inner prediction;
- BEST-to-GALLEX/SAGE transfer;
- GALLEX/SAGE-to-BEST transfer;
- no post-hoc tuning after seeing target data.

2.7 External vetoes

The active framework must survive quantitative veto checks:

- KATRIN energy-response constraints;
- IceCube / SME-style anisotropy or Lorentz-violation constraints where relevant;
- solar-neutrino consistency;
- standard three-flavor oscillation fits;
- detector calibration and source cross-section constraints.

These are constraints, not optional discussion points.

3. Active Canonical Dataset Scope

BEST

Active role:

- primary two-zone gallium source target;
- inner and outer zone ratio structure;
- future BEST-2 freeze target.

Current limitation:

- public data are insufficient for full likelihood-level validation without covariance, extraction-level data and metadata.

GALLEX

Active role:

- cross-experiment validation target;
- source-calibration comparison.

Current limitation:

- exact source coordinates, exposure-level tables and full response covariance are missing from current local package.

SAGE

Active role:

- cross-experiment validation target;
- isotope comparison using ^{51}Cr and ^{37}Ar source calibrations.

Current limitation:

- run-by-run exposure, impurity and counter-level likelihood data are missing.

BEST-2

Active role:

- future discriminating validation environment.

Required metadata:

- time-stamped extraction and exposure data;
- source-position and source-extension model;
- zone-by-zone covariance matrix;
- detector orientation metadata;
- environmental/systematic monitoring;
- public baseline prediction files.

4. Active Parameter Structures

The active parameter structures are limited to those needed by the 6.3.0 validation template:

- λ_0 : scalar trace response amplitude;
- λ_a : anisotropic response amplitude;
- A_a^{ij} : symmetric traceless anisotropy tensor for analysis bin or detector zone a ;
- $f_0(E, t)$: scalar response template, not yet physically fixed;
- $fA_a(E, t)$: anisotropic response template, not yet physically fixed;
- η : nuisance parameters for source strength, cross-section, geometry, extraction efficiency and normalization;
- C : statistical and systematic covariance matrix.

Audit restriction:

- A_a^{ij} must not remain an unconstrained five-parameter fit if the available data cannot identify it.
- Tensor orientation must be pre-specified for sidereal claims.

- `f0_a` and `fA_a` require explicit functional forms before any fit can be called frozen.

5. Active Claim Boundary

Allowed claims:

- LSC is an unvalidated phenomenological framework.
- LSC 6.3.0 defines a conservative pre-BEST-2 validation protocol.
- Trace/traceless separation is a useful way to separate normalization-like and directional-like residual templates.
- Sterile-neutrino explanations remain viable comparison baselines.
- Detector systematics remain active competitors.
- BEST-2 requires preregistered templates and metadata.

Not allowed:

- LSC proves the gallium anomaly.
- LSC excludes sterile neutrinos.
- LSC tensor structures are confirmed physical fields.
- PBH/curvature mechanisms are part of the current Gallium core.
- Anchor fits validate LSC.
- Detector systematics can be dismissed without a quantitative M4 baseline.

6. Canonicalization Decision

The canonical core is the LSC 6.3.0 validation framework after audit constraints:

- keep trace/traceless response separation;
- keep M0-M6 baseline hierarchy;
- keep frozen prediction protocol;
- keep leave-one-out and cross-experiment validation requirements;
- keep external veto requirements;
- downgrade all physical interpretation stronger than detector-response phenomenology;
- archive PBH/curvature-heavy and redundant historical material.

LSC Active Equation Set

Status: Step 1 canonicalization output

Rule: no new equations are introduced. The active set is restricted to equations already present in the LSC 6.3.0 / master package sources or directly classified in the external audits as freeze-core candidates.

1. Classification Key

- canonical: active in the pre-BEST-2 freeze framework.

- dimensional review needed: active only with explicit caveat; cannot support strong claims until reviewed.
- exploratory: useful for model-building but not frozen as validation core.
- deprecated: excluded from active core.
- historical only: retained for lineage.

2. Canonical Equations

E1. Baseline event expectation

```

NO_a = integral dE dOmega dt [
  Phi(E,t,Omega)
  P_alpha_beta(E,L)
  sigma(E)
  epsilon0_a(E,t,Omega)
  W_a(E,t,Omega)
]

```

Classification: canonical

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- a: analysis bin or detector zone;
- Phi: source flux;
- P_alpha_beta: oscillation probability or survival/transition factor;
- sigma: capture cross section;
- epsilon0_a: baseline detector efficiency;
- W_a: geometry/exposure weighting.

Physical meaning:

- defines the non-LSC baseline expectation.

Validation relevance:

- required to compare LSC to null and baseline models.

Dataset tie:

- BEST/GALLEX/SAGE rate predictions; needs source lines, geometry, efficiencies and covariance.

E2. LSC detector-efficiency response

```

epsilon_LSC,a(E,t,Omega)
= epsilon0_a(E,t,Omega) *
  [1 + lambda0 * f0_a(E,t) + lambdaA * q_a(t,Omega) * fA_a(E,t)]

```

Classification: canonical with functional-form freeze required

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- **lambda0**: scalar trace-response amplitude;
- **lambdaA**: anisotropic response amplitude;
- **f0_a**: scalar response template;
- **fA_a**: anisotropic response template;
- **q_a**: anisotropic projection.

Physical meaning:

- active detector-response ansatz.

Validation relevance:

- separates trace-only and trace-plus-anisotropy models.

Dataset tie:

- BEST/BEST-2 zone data; GALLEX/SAGE transfer only if geometry and metadata are available.

Audit caveat:

- **f0_a** and **fA_a** must be explicitly frozen before fitting.

E3. Anisotropic projection

$$q_a(t, \Omega) = n_i^{\text{lab}}(t, \Omega) A_a^{\{ij\}} n_j^{\text{lab}}(t, \Omega)$$

Classification: canonical with identifiability caveat

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- **n_i^{lab}**: lab-frame direction vector;
- **A_a^{ij}**: anisotropy tensor for bin/zone **a**.

Physical meaning:

- projects directional response into the detector/lab frame.

Validation relevance:

- needed for traceless anisotropy and sidereal/orientation tests.

Dataset tie:

- requires orientation/timestamp metadata; currently underconstrained.

Audit caveat:

- a freely fitted tensor is overparameterized with current public data.

E4. Tensor symmetry and tracelessness

$$A_a^{\{ij\}} = A_a^{\{ji\}} \\ \text{Tr}(A_a) = 0$$

Classification: canonical

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- $A_a^{\{ij\}}$: symmetric traceless response tensor.

Physical meaning:

- removes scalar normalization from directional component.

Validation relevance:

- prevents trace response and anisotropy from being collapsed into one undifferentiated fit term.

Dataset tie:

- only testable with enough directional/orientation information.

E5. LSC event expectation

$$N_{\text{LSC},a}(\theta) = \\ \text{integral } dE \, d\Omega \, dt \, [\\ \quad \Phi \, P \, \sigma \, \epsilon_{\text{LSC},a} \, W_a \\]$$

Classification: canonical

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- θ : LSC parameters and nuisance parameters as specified in the frozen model.

Physical meaning:

- event expectation after applying the LSC detector-response template.

Validation relevance:

- direct model-comparison object.

Dataset tie:

- requires same source/geometry/covariance inputs as E1.

E6. Observed ratio

$$R_a = N_{\text{obs},a} / N0_a$$

Classification: canonical

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- $N_{\text{obs},a}$: observed count/rate;
- $N0_a$: baseline expected count/rate.

Physical meaning:

- rate-ratio residual target.

Validation relevance:

- basic input to BEST/GALLEX/SAGE comparison.

Dataset tie:

- available in summary form; full likelihood requires missing details.

E7. First-order LSC residual template

$$\Delta R_a^{\text{LSC}} \sim \lambda_{00} * F0_a + \lambda_{0A} * FA_a$$

Classification: canonical as first-order template

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- $F0_a$: scalar weighted response;
- FA_a : anisotropic weighted response.

Physical meaning:

- linearized residual decomposition.

Validation relevance:

- supports M5 trace-only and M6 trace-plus-anisotropy comparison.

Dataset tie:

- requires weighting definitions and covariance.

E8. Weighted scalar and anisotropic factors

$$F0_a = \langle f0_a \rangle_a$$

$$FA_a = \langle q_a fA_a \rangle_a$$

Classification: canonical with weighting-definition caveat

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- brackets denote the same flux, cross-section, geometry and exposure weighting used in NO_a.

Physical meaning:

- converts functional templates into bin-level residual factors.

Validation relevance:

- required to avoid free post-hoc fitting of bin coefficients.

Dataset tie:

- needs explicit weights from experiment geometry and exposure.

E9. BEST binned target

$$R_{\text{BEST}} = (R_{\text{inner}}, R_{\text{outer}}, R_{\text{outer}} / R_{\text{inner}})$$

Classification: canonical

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- **R_inner**: inner-zone BEST ratio;
- **R_outer**: outer-zone BEST ratio.

Physical meaning:

- minimum BEST vector for two-zone tests.

Validation relevance:

- necessary but insufficient; cannot validate anisotropy alone.

Dataset tie:

- BEST summary data; full covariance still missing.

E10. Covariance-aware validation statistic

$$\begin{aligned} \text{chi2}(\text{theta}, \text{eta}) = & \\ & [R_{\text{obs}} - R_{\text{pred}}(\text{theta}, \text{eta})]^T \\ & C^{-1} \\ & [R_{\text{obs}} - R_{\text{pred}}(\text{theta}, \text{eta})] \\ & + \text{penalty}(\text{eta}) \end{aligned}$$

Classification: canonical with missing-covariance caveat

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- **theta**: LSC parameters;
- **eta**: nuisance parameters;
- **C**: statistical and systematic covariance matrix;

- `penalty(eta)`: model/nuisance penalty.

Physical meaning:

- formal comparison statistic.

Validation relevance:

- mandatory for meaningful model comparison.

Dataset tie:

- currently blocked by missing full covariance matrices.

E11. Frame transformation for sidereal/orientation tests

`n_lab(t) = R_lab<-ICRS(t, latitude, longitude, detector_orientation) * n_ICRS`

Classification: canonical for future BEST-2 tests; currently metadata-blocked

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- `n_ICRS`: fixed celestial-frame direction;
- `R_lab<-ICRS`: time-dependent transform to lab frame.

Physical meaning:

- distinguishes lab-fixed detector artifacts from celestial-frame anisotropy.

Validation relevance:

- essential for sidereal claims.

Dataset tie:

- requires timestamp, latitude/longitude and detector orientation metadata.

E12. Sidereal template

`S_a(t) = n_lab,i(t) A^{ij} n_lab,j(t)`

Classification: canonical for future test; speculative until metadata exists

Source: LSC 6.3.0 Unified BEST-2 Continuation

Variables:

- `S_a(t)`: time-dependent anisotropy signal;
- `A^{ij}`: traceless anisotropy tensor.

Physical meaning:

- predicted sidereal/orientation modulation template.

Validation relevance:

- strongest proposed discriminator, but not currently testable with public data.

Dataset tie:

- BEST-2 metadata required.

3. Active But Dimensional Review Needed

DR1. Detector-frame energy reconstruction form

$$E_{\text{rec}} = E_{\text{true}} * [1 + \alpha_D * D_{ij} * p_{\text{hat}}^i * p_{\text{hat}}^j]$$

Classification: dimensional review needed

Source: LSC 6.2.0 and historical 6.2.x line

Reason:

- audits flag tensor dimensions, identifiability and possible Lorentz-violation implications.
- current 6.3.0 operational core can be expressed through detector efficiency response instead.

Validation relevance:

- not sufficient as a validation equation unless mapped to E2/E5 and external vetoes.

Dataset tie:

- indirect only.

DR2. Effective propagation factor

$$G(\Phi, E) = 1 + \delta G(\Phi, E) + O(\delta G^2)$$

Classification: dimensional review needed / effective-only

Source: LSC 6.0 / 6.2.0 lineage

Reason:

- δG has no accepted microscopic derivation in the current freeze.
- curvature/gravitational labels must be removed from active claims.

Validation relevance:

- can remain as effective nuisance/response language only.

Dataset tie:

- not directly identifiable with current Gallium summary data.

4. Deprecated Equation Families

DEP1. PBH/curvature Hamiltonian family

Classification: deprecated / historical only

Includes:

- explicit PBH/curvature H_{LSC} ;
- explicit gravitational Hamiltonian terms as current Gallium mechanism;
- PBH resonance equations.

Reason:

- removed from current Gallium core;
- no direct validation path in BEST/GALLEX/SAGE.

DEP2. Schwarzschild energy-map family

Classification: deprecated / historical only

Includes:

- old E_{obs} / E_{emit} Schwarzschild-style mapping used as standalone explanation.

Reason:

- replaced by effective detector-response framing.

DEP3. Covariant Lagrangian from LSC 5.5

Classification: deprecated / historical only

Reason:

- Lorentz-violation and SME implications are unresolved;
- not used in current BEST-2 validation protocol.

DEP4. Redundant tensor parameterizations

Classification: deprecated unless reduced to the single 6.3.0 trace/traceless structure

Reason:

- multiple equivalent tensor forms create ambiguity and overfitting risk.

5. Equation Freeze Decision

Frozen as active:

- E1-E12 above.

Active only with caveats:

- DR1 and DR2.

Archived:

- DEP1-DEP4 and all historical equation clutter not needed by the 6.3.0 validation protocol.

LSC Freeze Boundary

Status: Step 1 canonicalization output

Boundary: pre-BEST-2 freeze era

1. What Is Frozen

The following are frozen:

- LSC status as an unvalidated phenomenological validation framework.
- LSC 6.3.0 as the current freeze state.
- Zenodo concept DOI 10.5281/zenodo.19780615 as primary identity.
- Trace/traceless detector-response separation.
- M0-M6 baseline hierarchy.
- Requirement that detector systematics remain an active competitor.
- Requirement that sterile-neutrino baselines remain active.
- Requirement for leave-one-out and cross-experiment validation.
- Requirement for covariance-aware model comparison.
- Requirement for quantitative external veto checks.
- Claim boundary: no discovery, no sterile exclusion, no confirmed tensor field.

2. What Is Not Frozen

The following are not frozen because they remain underdefined or data-blocked:

- exact functional forms for $f0_a(E,t)$ and $fA_a(E,t)$;
- physically motivated tensor orientation;
- final parameter bounds;
- full detector-systematics baseline M4;
- full covariance model;
- KATRIN/IceCube/solar quantitative veto mapping;
- BEST-2 prediction values;
- sidereal modulation amplitude and phase.

These must not be retroactively fitted after BEST-2 data are seen.

3. Prohibited Future Extension Without New Data

Without new data or explicit preregistration, it is prohibited to:

- add new mechanisms;
- add new tensor structures;
- add new particle claims;
- revive PBH/curvature language as active core;
- reinterpret detector systematics as LSC evidence without M4 comparison;
- tune tensor orientation after data inspection;
- claim sidereal evidence without timestamps and detector orientation;
- claim statistical preference without full covariance or a clearly labeled exploratory approximation;
- treat AI audit agreement as physics evidence.

4. Allowed Claims

Allowed:

- LSC 6.3.0 is a conservative pre-BEST-2 validation framework.
- The framework tests trace and traceless detector-response templates against required baselines.
- Current public data are insufficient for full likelihood-level validation.
- Full covariance and metadata are required before strong claims.
- Negative results are valid freeze-era outcomes.

5. Not Allowed Claims

Not allowed:

- LSC explains the Gallium anomaly.
- LSC proves new physics.
- LSC excludes sterile neutrinos.
- LSC tensor parameters are confirmed physical fields.
- LSC has survived KATRIN/IceCube/solar vetoes unless explicitly computed.
- Anchor fits validate the theory.
- BEST-2 will confirm LSC.

6. Falsification or Weakening Conditions

LSC is weakened if:

- M4 detector-systematics baseline reproduces all LSC-like structures;
- M1/M2 normalization or cross-section nuisance models perform as well as M5/M6 after penalties;
- leave-one-out remains unstable;
- cross-experiment transfer fails;
- anisotropy collapses into scalar normalization under covariance propagation;
- each experiment requires unrelated tensor parameters;

- external vetoes exclude the required parameter region.

LSC requires major revision or abandonment if:

- frozen templates fail against BEST-2 while simpler baselines succeed;
- sidereal/orientation predictions fail where required metadata exist;
- the model cannot be distinguished from mundane detector effects;
- external vetoes are quantitatively incompatible with required amplitudes.

7. Structures Remaining Unresolved

Unresolved:

- physical origin of `delta_G` or equivalent effective response term;
- physical status of the anisotropy tensor;
- Lorentz-violation / SME mapping;
- KATRIN energy-response mapping;
- IceCube constraint mapping;
- detector-systematics M4 implementation;
- covariance access;
- BEST-2 metadata access;
- identifiability of trace vs traceless terms with realistic data.

8. Step 2 Requirements

Step 2 must not expand the theory. It must build validation infrastructure:

- define exact data requirements;
- create covariance placeholder policy;
- define M4 detector-systematics baseline specification;
- freeze `f0_a` and `fA_a` only if source-justified;
- define leave-one-out and cross-experiment validation protocol;
- define external veto calculation checklist;
- produce a preregistration-ready validation plan.

LSC Historical Lineage

Status: Step 1 canonicalization output

Purpose: clean lineage without importing deprecated material into the active core

1. Early LSC / SC neutrino line

Core idea:

- early exploration of neutrino anomalies, propagation effects and speculative mechanisms.

What survived:

- the question of whether Gallium/BEST residuals can be modeled without assuming sterile neutrinos as the only explanation.

What was removed:

- non-observable language;
- non-scientific or AI/meta-analysis framing;
- unsupported ontological claims.

Canonical result:

- only the neutrino phenomenology question survived.

Deprecated result:

- speculative or non-physics framing.

2. LSC 4.x / LSC 4.2 ULTRA

Core idea:

- gravitationally coupled neutrino oscillation framework;
- PBH and curvature-centered language;
- effective Hamiltonian and resonance concepts.

What survived:

- historical recognition that LSC began with propagation-side modifications.
- the later need to define observables and validation constraints.

What was removed:

- PBH-centered Gallium explanation;
- compact-object phenomenology as active core;
- curvature-heavy physical interpretation for percent-level laboratory deficits.

Canonical result:

- none of the PBH/curvature mechanism is active in the pre-BEST-2 core.

Deprecated result:

- PBH/curvature mechanism and explicit gravitational operator language.

3. LSC 5.x / LSC 5.0@

Core idea:

- neutrino anomalies may be related to energy reconstruction rather than only new particles;
- E_{true} / E_{obs} distinction became central.

What survived:

- detector/reconstruction framing as a historical bridge.

What was removed:

- Schwarzschild-style energy mapping as standalone explanation;
- strong claims that relativistic energy shifts reproduce Gallium deficits.

Canonical result:

- later detector-response language, not the old physical mechanism.

Deprecated result:

- standalone relativistic energy-map interpretation.

4. LSC 5.5

Core idea:

- combine propagation modulation with nonlinear tensor energy reconstruction;
- introduce tensor-style detector response.

What survived:

- the idea that response can have scalar and directional components.

What was removed:

- broad resonant geometric modulation language;
- covariant Lagrangian as current theory;
- redundant tensor parameterizations;
- unsupported physical interpretation of the tensor.

Canonical result:

- only the restricted trace/traceless detector-response split survived into 6.3.0.

Deprecated result:

- unconstrained tensor freedom and 5.5 physical overreach.

5. LSC 6.0

Core idea:

- clean phenomenological framework for propagation factor plus detector response;
- remove compact-object assumptions from Gallium interpretation;
- define external constraints such as KATRIN and IceCube as necessary checks.

What survived:

- conservative phenomenological status;
- detector response tensor as effective object;
- need for Gallium/BEST, KATRIN and IceCube consistency checks.

What was removed:

- strong curvature/PBH claims;
- direct curvature causation language.

Canonical result:

- scientific posture: proposal -> numerical tests -> constraints -> refinement.

Deprecated result:

- residual curvature labels where not operational.

6. LSC v1.2.0 computational supplement

Core idea:

- computational modeling and reproducibility update.

What survived:

- reproducibility and source-package role.

What was removed:

- treatment as a separate primary theory line.

Canonical result:

- supplement only.

Deprecated result:

- any interpretation that v1.2.0 supersedes the main LSC publication line.

7. LSC 6.2.x

Core idea:

- detector-frame tensor anisotropy;
- explicit energy reconstruction and oscillation phase language;
- later correction of formal issues.

What survived:

- detector-frame anisotropy as a test class;
- trace/traceless correction path;
- leave-one-out warning;
- dimensional and frame-consistency discipline.

What was removed:

- excessive equation inventory;
- anchor-fit validation confidence;
- underdefined tensor freedom;
- unreviewed physical interpretations.

Canonical result:

- LSC must be validated by leave-one-out, cross-experiment transfer and baseline comparison.

Deprecated result:

- treating 6.2.0 preprint language as final without 6.2.1/6.2.2 corrections.

8. LSC 6.3.0

Core idea:

- unified BEST-2 continuation;
- scalar trace vs traceless anisotropy separation;
- frozen prediction protocol;
- M0-M6 baseline hierarchy;
- external vetoes;
- negative results archived as valid outcomes.

What survived:

- active canonical core.

What was removed:

- discovery language;
- sterile-neutrino exclusion language;
- AI-model agreement as evidence;
- full EFT/covariant/PMNS completion claims.

Canonical result:

- the current pre-BEST-2 freeze core.

Deprecated result:

- anything stronger than constrained phenomenological validation.